

## DANVILLE, VA, USA

WMO: 724106

Lat: 36.573N

Lon: 79.335W

Elev: 571

StdP: 14.40

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13728

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.0       | 21.6 | 0.3                           | 5.7 | 25.7 | 4.8 | 7.2 | 28.1 | 22.2                  | 45.7 | 19.4 | 45.8 | 4.8                   | 260  | 0.412 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.5                   | 93.5            | 74.5 | 91.2 | 74.3 | 89.4 | 73.7 | 77.9                | 88.6 | 76.8 | 86.9 | 75.7 | 85.0 | 7.2                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.0                            | 134.0 | 82.3 | 73.7 | 128.3 | 80.8 | 72.9 | 124.5 | 79.9 | 41.6          | 88.3 | 40.5 | 87.4 | 39.4 | 85.3 | 84.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| 18.6              | 16.3 | 13.8 | 10.6                       | 97.1 | 5.3                | 3.2 | 6.8                                                | 99.3 | 3.7        | 101.2 | 0.7        | 102.9 | -3.1       | 105.2 |    |    |
|                   |      |      | 9.2                        | 80.5 | 5.0                | 1.8 | 5.6                                                | 81.8 | 2.7        | 82.9  | -0.1       | 83.9  | -3.7       | 85.2  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 59.1   | 39.5  | 42.3  | 49.4  | 58.8  | 67.1  | 74.6  | 78.0  | 77.1  | 70.6  | 59.7  | 48.9  | 42.1  |
|                                                                        | DBStd       | 15.74  | 9.91  | 9.30  | 9.54  | 8.16  | 6.95  | 5.20  | 4.12  | 4.40  | 6.02  | 7.94  | 8.47  | 8.87  |
|                                                                        | HDD50       | 1164   | 352   | 250   | 132   | 18    | 0     | 0     | 0     | 0     | 0     | 13    | 121   | 278   |
|                                                                        | HDD65       | 3618   | 790   | 637   | 490   | 218   | 63    | 4     | 0     | 0     | 20    | 201   | 484   | 711   |
|                                                                        | CDD50       | 4485   | 27    | 33    | 112   | 283   | 530   | 737   | 869   | 840   | 618   | 314   | 89    | 33    |
|                                                                        | CDD65       | 1466   | 0     | 0     | 6     | 33    | 128   | 291   | 405   | 375   | 188   | 37    | 2     | 1     |
|                                                                        | CDH74       | 13295  | 2     | 6     | 70    | 404   | 1170  | 2679  | 3750  | 3365  | 1477  | 343   | 24    | 5     |
|                                                                        | CDH80       | 5226   | 0     | 0     | 7     | 90    | 357   | 1074  | 1652  | 1443  | 523   | 79    | 1     | 0     |
| (14) Wind                                                              | WSAvg       | 5.7    | 6.5   | 6.6   | 7.1   | 7.1   | 5.7   | 5.1   | 4.8   | 4.4   | 5.0   | 5.1   | 5.5   | 5.6   |
| (15) Precipitation                                                     | PrecAvg     | 43.1   | 3.6   | 3.3   | 4.0   | 3.4   | 3.9   | 3.7   | 4.3   | 3.8   | 3.8   | 3.3   | 3.0   | 3.1   |
|                                                                        | PrecMax     | 63.3   | 8.7   | 6.6   | 10.4  | 8.3   | 7.7   | 14.6  | 8.8   | 9.6   | 14.6  | 10.9  | 7.6   | 6.4   |
|                                                                        | PrecMin     | 28.7   | 0.7   | 0.4   | 0.7   | 0.3   | 1.1   | 1.1   | 1.6   | 0.9   | 0.2   | 0.1   | 0.3   | 0.3   |
|                                                                        | PrecStd     | 7.5    | 1.8   | 1.5   | 2.1   | 2.0   | 1.8   | 2.7   | 1.6   | 2.0   | 3.1   | 2.4   | 1.7   | 1.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 71.4  | 73.4  | 80.5  | 87.5  | 90.5  | 95.0  | 97.4  | 96.7  | 93.0  | 86.8  | 77.4  | 72.7  |
|                                                                        |             | MCWB   | 60.4  | 59.8  | 61.3  | 65.9  | 72.2  | 74.9  | 75.8  | 74.7  | 71.5  | 71.1  | 62.2  | 60.4  |
|                                                                        | 2%          | DB     | 65.2  | 68.3  | 75.3  | 82.2  | 87.2  | 91.4  | 93.4  | 93.2  | 89.5  | 81.5  | 72.4  | 66.2  |
|                                                                        |             | MCWB   | 58.0  | 56.8  | 59.0  | 63.9  | 70.7  | 74.3  | 75.3  | 74.8  | 71.9  | 66.9  | 60.2  | 59.3  |
|                                                                        | 5%          | DB     | 61.2  | 63.6  | 71.2  | 79.1  | 84.1  | 89.2  | 91.0  | 90.5  | 86.1  | 77.7  | 68.4  | 62.0  |
|                                                                        |             | MCWB   | 54.0  | 53.4  | 56.5  | 62.2  | 69.8  | 73.5  | 75.2  | 74.4  | 70.8  | 65.4  | 58.8  | 55.7  |
|                                                                        | 10%         | DB     | 55.7  | 58.7  | 66.2  | 74.7  | 81.1  | 86.4  | 88.5  | 87.9  | 82.3  | 73.5  | 64.3  | 57.0  |
|                                                                        |             | MCWB   | 48.5  | 49.1  | 54.4  | 60.4  | 67.9  | 72.4  | 74.5  | 73.7  | 69.5  | 64.0  | 56.5  | 50.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 63.6  | 64.1  | 64.7  | 69.5  | 75.1  | 77.9  | 79.4  | 79.2  | 76.7  | 74.2  | 67.3  | 65.2  |
|                                                                        |             | MCDB   | 66.9  | 69.7  | 74.5  | 77.5  | 84.8  | 89.0  | 90.7  | 89.9  | 85.1  | 79.5  | 71.7  | 70.6  |
|                                                                        | 2%          | WB     | 59.6  | 60.0  | 62.2  | 67.0  | 73.1  | 76.4  | 78.1  | 77.6  | 74.8  | 71.1  | 64.2  | 61.1  |
|                                                                        |             | MCDB   | 64.6  | 66.6  | 70.9  | 77.4  | 82.9  | 87.3  | 89.1  | 88.0  | 83.2  | 77.7  | 68.4  | 64.3  |
|                                                                        | 5%          | WB     | 55.3  | 55.3  | 59.8  | 64.9  | 71.4  | 75.2  | 77.1  | 76.4  | 73.2  | 68.6  | 61.3  | 56.6  |
|                                                                        |             | MCDB   | 59.4  | 60.9  | 67.7  | 74.8  | 81.1  | 85.4  | 87.5  | 85.9  | 81.2  | 74.7  | 67.0  | 60.9  |
|                                                                        | 10%         | WB     | 49.5  | 50.4  | 56.8  | 62.8  | 69.6  | 74.0  | 75.9  | 75.2  | 71.7  | 65.8  | 57.7  | 51.4  |
|                                                                        |             | MCDB   | 54.6  | 57.0  | 64.2  | 71.6  | 77.6  | 83.5  | 85.4  | 83.9  | 78.8  | 71.6  | 62.7  | 55.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.2  | 20.7  | 22.3  | 23.5  | 21.6  | 20.5  | 19.5  | 19.7  | 20.1  | 22.5  | 22.4  | 19.1  |
|                                                                        |             | MCDBR  | 25.3  | 27.4  | 28.1  | 28.5  | 25.2  | 23.6  | 22.9  | 23.4  | 25.0  | 26.4  | 27.2  | 24.3  |
|                                                                        |             | MCWBR  | 17.8  | 18.2  | 15.1  | 13.3  | 11.4  | 9.3   | 8.3   | 8.1   | 10.0  | 13.5  | 16.6  | 17.3  |
|                                                                        | 5% WB       | MCDBR  | 22.4  | 23.3  | 23.0  | 23.3  | 21.3  | 20.7  | 19.9  | 19.7  | 19.3  | 20.3  | 20.7  | 20.1  |
|                                                                        |             | MCWBR  | 18.2  | 18.2  | 14.7  | 12.4  | 10.6  | 9.2   | 8.3   | 7.9   | 8.7   | 11.7  | 15.6  | 16.4  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.303 | 0.322 | 0.352 | 0.395 | 0.435 | 0.481 | 0.518 | 0.517 | 0.408 | 0.367 | 0.322 | 0.307 |
|                                                                        | taud        |        | 2.511 | 2.456 | 2.414 | 2.304 | 2.214 | 2.128 | 2.047 | 2.027 | 2.330 | 2.428 | 2.514 | 2.535 |
|                                                                        | Ebn at Noon |        | 284   | 291   | 290   | 282   | 271   | 257   | 246   | 243   | 268   | 270   | 274   | 275   |
|                                                                        | Edh at Noon |        | 25    | 30    | 35    | 41    | 45    | 49    | 53    | 53    | 37    | 31    | 25    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 788   | 1006  | 1332  | 1669  | 1849  | 2005  | 1870  | 1704  | 1404  | 1143  | 865   | 674   |
|                                                                        | RadStd      |        | 53    | 121   | 92    | 133   | 153   | 145   | 123   | 88    | 133   | 141   | 67    | 58    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (53 neighbors) |       | +0.68   | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +230  | +153  |

Nomenclature: See separate page